

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q20-1390

Mclain 161 kV  
May 20, 2021

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1390. The Interconnection Customer proposes a 89.4 MW solar photovoltaic facility interconnecting at the Mclain 161 kV substation in the New York area. The request was received on January 17, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 465 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$36,187,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: Hughes-Mclain 161.0 kV  
Contingency: Closter-Lee Vining 161.0 kV Line  
Loading: 126.0% (Rating: 276 MVA)  
DFAX: 54.1%  
Pre-project loading: 108.5%
2. Facility: Hughes-Wells Rver 161.0 kV  
Contingency: Lee Vining-Closter 161.0 kV  
Loading: 119.2% (Rating: 264 MVA)  
DFAX: 54.1%  
Pre-project loading: 100.8%
3. Facility: Lee Vining-Loch Mirror 161.0 kV  
Contingency: Hughes-Mclain 161.0 kV  
Loading: 116.6% (Rating: 344 MVA)  
DFAX: 47.3%  
Pre-project loading: 104.3%
4. Facility: Closter-Lee Vining Substation 161.0 kV  
Contingency: Hughes-Mclain 161.0 kV  
Loading: 114.7% (Rating: 370 MVA)  
DFAX: 47.3%  
Pre-project loading: 103.3%
5. Facility: Wells River-Gastonia Prime Power Park 161.0 kV  
Contingency: Lee Vining-Closter 161.0 kV  
Loading: 111.9% (Rating: 280 MVA)  
DFAX: 43.2%  
Pre-project loading: 98.2%

6. Facility: Loch Mirror 161 kV/138 kV TX  
 Contingency: Gastonia Prime Power Park-Salton Sea 2 138.0 kV  
 Loading: 107.0% (Rating: 150 MVA)  
 DFAX: 5.2%  
 Pre-project loading: 104.0%
7. Facility: Loch Mirror 161/138kV Transformer Bank T2  
 Contingency: Salton Sea 2-Gastonia Prime Power Park 138.0 kV  
 Loading: 107.0% (Rating: 150 MVA)  
 DFAX: 5.2%  
 Pre-project loading: 104.0%
8. Facility: Loch Miror 161 kV/138 kV Transformer  
 Contingency: Gastonia Prime Power Park-Salton Sea 2 138.0 kV L  
 Loading: 107.0% (Rating: 150 MVA)  
 DFAX: 5.2%  
 Pre-project loading: 104.0%
9. Facility: Loch Mirror 161-138 kV XFMR Bank 4  
 Contingency: Salton Sea 2-Gastonia Prime Power Park Substation 138.0 kV  
 Loading: 107.0% (Rating: 150 MVA)  
 DFAX: 5.2%  
 Pre-project loading: 104.0%

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                                  | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Hughes-Mclain 161.0 kV                                    | Rebuild line with higher-capacity conductor | \$2,151,000   | 100% (\$2,151,000)    |
| Hughes-Wells Rver 161.0 kV                                | Rebuild line with higher-capacity conductor | \$1,416,000   | 100% (\$1,416,000)    |
| Lee Vining-Loch Mirror 161.0 kV                           | Rebuild line with higher-capacity conductor | \$2,191,000   | 100% (\$2,191,000)    |
| Closter-Lee Vining Substation 161.0 kV                    | Rebuild line with higher-capacity conductor | \$4,333,000   | 100% (\$4,333,000)    |
| Wells River Substation-Gastonia Prime Power Park 161.0 kV | Rebuild line with higher-capacity conductor | \$804,000     | 100% (\$804,000)      |
| Loch Mirror 161-138 kV Auto T1                            | Replace transformer with higher-rated unit  | \$6,688,000   | 100% (\$6,688,000)    |
| Loch Mirror 161 kV/138 kV TX                              | Replace transformer with higher-rated unit  | \$9,040,000   | 100% (\$9,040,000)    |
| Loch Mirror 161 kV/138 kV Autotransformer                 | Replace transformer with higher-rated unit  | \$5,044,000   | 100% (\$5,044,000)    |
| Loch Mirror 161-138 kV XFMR Bank 4                        | Replace transformer with higher-rated unit  | \$4,520,000   | 100% (\$4,520,000)    |
|                                                           | Total allocated to project                  |               | \$36,187,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.